

Student Name	MONISHA S
Register Number	TU6253210211034
Project Title	Seller and Inventory Management System
Week	Week 3

WEEK-3 TASK

Task Name

Seller and Inventory Management System

Abstract

The Seller and Inventory Management System is a database management system designed to manage seller details, product information, and inventory records efficiently. The system maintains relationships between sellers and products, tracks stock availability, and generates inventory status reports. It helps organizations monitor product stock levels, identify unavailable items, and ensure effective inventory control.

Objectives

- Create Seller and Inventory tables.
- Establish relationships between sellers, products, and stock.
- Maintain seller product information.
- Track available and unavailable products.
- Generate inventory status reports.

Database Design

Seller Table

Column Name	Data Type	Description
Seller_ID	INT	Primary Key
Seller_Name	VARCHAR(50)	Seller Name
Contact_No	VARCHAR(15)	Contact Number
City	VARCHAR(50)	Seller Location

Product Table

Column Name	Data Type	Description
Product_ID	INT	Primary Key
Product_Name	VARCHAR(50)	Product Name
Price	DECIMAL(10,2)	Product Price
Seller_ID	INT	Foreign Key

Inventory Table

Column Name	Data Type	Description
Inventory_ID	INT	Primary Key
Product_ID	INT	Foreign Key
Stock_Quantity	INT	Available Quantity
Status	VARCHAR(20)	Available/Unavailable

ER Diagram

SELLER

Seller_ID (PK)

Seller_Name

Contact_No

City

|

| 1

|

| M

PRODUCT

Product_ID (PK)

Product_Name

Price

Seller_ID (FK)

|

| 1

|

| 1

INVENTORY

Inventory_ID (PK)

Product_ID (FK)

Stock_Quantity

Status

SQL Code

```
CREATE DATABASE SellerInventoryDB;
```

```
USE SellerInventoryDB;
```

```
-- Seller Table
```

```
CREATE TABLE Seller (
```

```
    Seller_ID INT PRIMARY KEY,
```

```
    Seller_Name VARCHAR(50),
```

```
    Contact_No VARCHAR(15),
```

```
    City VARCHAR(50)
```

```
);
```

```
-- Product Table
```

```
CREATE TABLE Product (
```

```
    Product_ID INT PRIMARY KEY,
```

```
    Product_Name VARCHAR(50),
```

```
    Price DECIMAL(10,2),
```

```
    Seller_ID INT,
```

```
    FOREIGN KEY (Seller_ID)
```

```
    REFERENCES Seller(Seller_ID)
```

```
);
```

```
-- Inventory Table
```

```

CREATE TABLE Inventory (
    Inventory_ID INT PRIMARY KEY,
    Product_ID INT,
    Stock_Quantity INT,
    Status VARCHAR(20),
    FOREIGN KEY (Product_ID)
    REFERENCES Product(Product_ID)
);

-- Insert Seller Records
INSERT INTO Seller VALUES
(101,'Ramesh','9876543210','Chennai'),
(102,'Suresh','9876543211','Coimbatore'),
(103,'Priya','9876543212','Madurai');

-- Insert Product Records
INSERT INTO Product VALUES
(201,'Laptop',55000,101),
(202,'Mobile',25000,102),
(203,'Headphones',2000,103);

-- Insert Inventory Records
INSERT INTO Inventory VALUES
(301,201,20,'Available'),
(302,202,0,'Unavailable'),
(303,203,50,'Available');

```

Queries

1. Display Seller and Product Details

```

SELECT S.Seller_Name,
       P.Product_Name,
       P.Price

```

FROM Seller S

JOIN Product P

ON S.Seller_ID = P.Seller_ID;

2. Display Available Products

SELECT P.Product_Name,

I.Stock_Quantity

FROM Product P

JOIN Inventory I

ON P.Product_ID = I.Product_ID

WHERE I.Status='Available';

3. Display Unavailable Products

SELECT P.Product_Name

FROM Product P

JOIN Inventory I

ON P.Product_ID = I.Product_ID

WHERE I.Status='Unavailable';

4. Inventory Status Report

SELECT P.Product_Name,

I.Stock_Quantity,

I.Status

FROM Product P

JOIN Inventory I

ON P.Product_ID = I.Product_ID;

Sample Output

Product Name	Stock Quantity	Status
Laptop	20	Available
Mobile	0	Unavailable
Headphones	50	Available

Conclusion

The Seller and Inventory Management System effectively manages seller information, product records, and inventory status. By maintaining relationships between sellers, products, and stock, the system helps track available and unavailable products and generates inventory reports for better inventory management and decision-making.